HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and deployed **Java Spring Boot** microservices for backend systems, implementing RESTful APIs and ensuring robust data exchange with upstream and downstream applications.
- Designed and implemented data persistence layers using **Hibernate** and JPA, managing complex object-relational mappings and optimizing database interactions for enterprise applications.
- Authored **JUnit** and **Mockito** tests for Java components, achieving over 85% code coverage and ensuring the reliability and maintainability of core application logic.
- Analyzed and modified **automation control system software** for validated manufacturing environments, ensuring compliance with strict medical device standards and documentation requirements.
- Participated in **cross-functional teams** to deliver automation projects on time and to required quality standards, contributing to the successful integration of new control systems on production lines.
- Provided **technical support to Operations** for automation-related issues on production lines, troubleshooting and resolving critical incidents to minimize manufacturing downtime.
- Containerized applications using **Docker** and deployed to **AWS EC2** instances, managing CI/CD pipelines with **Jenkins** for automated builds and deployments.

SKILLS

Programming Languages – Java, Python, SQL, R

Frameworks & Libraries – Spring Boot, Hibernate, JPA, JUnit, Mockito, pandas, NumPy, scikit-learn, Flask, FastAPI

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle

Tools & Monitoring – Splunk, Dynatrace, Jira, Confluence

Data Analytics & BI – Power BI, Tableau, DAX, Power Query, RStudio

Engineering Domains – Automation Engineering, Control Systems, Manufacturing Systems, Software Development

PROJECTS

Automated Manufacturing Control System Upgrade

- Developed and tested new **control system software** in **Java** and **Python** for a validated manufacturing line, integrating with existing sensor networks and actuators.
- Implemented comprehensive logging and error handling within the automation software, ensuring detailed audit trails and facilitating rapid fault diagnosis for production issues.
- Created **Docker** containers for the control system components, streamlining deployment to **AWS** environments and enabling consistent testing across development and staging.

Production Data Pipeline and Dashboard

- Engineered a **Python** ETL pipeline to extract, transform, and load manufacturing sensor data from operational systems into a central data store, processing approximately 15,000 data points per hour.
- Developed interactive **Power BI** dashboards to visualize key production metrics, including yield rates, machine uptime, and defect trends, providing actionable insights to operations teams.
- Wrote complex **SQL** queries against PostgreSQL to aggregate and validate data for reporting, ensuring data integrity for the analytical dashboards.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)